SAP CPI & API Management – Introduction

Standalone reference notes (not tied to a specific class day)

Verified against SAP's own documentation – accurate overall. One very minor polish: SAP's official lifecycle stage names are *Build API Proxies* → *Publish API Proxies* → *Consume APIs* → *Analyze API Proxies* (plus Secure and Govern as ongoing activities within those stages) – your 6-stage breakdown (Create, Secure, Publish, Consume, Monitor, Govern) captures the same ideas correctly, just with slightly more granular, easier-to-remember labels. Kept as you wrote it below, since it's a faithful and genuinely clearer way to learn the concept.

1. What is an API? (start here first)

API stands for **Application Programming Interface** – a set of rules and protocols that allows one application to communicate with another application.

Key points:

- An API acts as a **"messenger"** between applications (**Request** → **Process** → **Response**).
- Enables systems to **share data securely** between each other.
- **Common protocol:** HTTP / HTTPS.
- **Common style (methods):** GET, POST, PUT, DELETE.
- **Returns data** in **JSON** or **XML.**

How an API works

```
Application A (Client/Consumer) --Request (HTTP/HTTPS)--> API --Request/Response-->
Application B (Server/Provider) Application A <--Response-- API
```

Common example flow:

```
External Application (Consumer) -> SAP API Management (API Proxy) -> SAP CPI ->
SuccessFactors / S/4HANA (Backend System)
```

Real-time example

An external system (or application) wants **Employee data from SuccessFactors.** Instead of direct access to SuccessFactors, it calls an **API** instead.

Flow: Consumer sends a request for Employee data → **API Management** validates and forwards the request → **CPI** processes the request and calls the Backend System → the Backend System returns data, and a **Response** is sent back to the Consumer.

2. Why do organizations need API Management?

Without API Management:

```
Multiple Consumers --------> Backend Systems (direct, unmanaged access)
```

Challenges this creates:

- **No centralized security** (anyone can access).
- **No authentication or authorization control.**

- **No rate limiting** (risk of system overload).
- **No visibility** (hard to monitor API usage).
- **No standardization & governance.**
- **High maintenance** and governance overhead.
- **Security vulnerabilities** and compliance risk.

With API Management

It's a **centralized location** providing:

1. **Security** (Authentication & Authorization).
2. **Policies** (Rate limit, Quota, Validation).
3. **Traffic Control.**
4. **Monitor & Analytics.**
5. **Consumer Management.**
6. **API Lifecycle Management.**

```
Multiple Consumers -> [Security | Policies | Traffic Control | Monitor & Analytics | Consumer Mgmt | API Lifecycle Mgmt] -> Backend Systems
```

Bottom line: API Management acts as a **smart gateway** between consumers and backend systems – ensuring security, control, visibility, and a great experience for both business and developers.

What is API Management, really? It is **not** simply about exposing an API – it manages the **complete lifecycle**, from **create** to **governance.**

3. The API Management Lifecycle

Stage	What happens
1. Create	Design & develop the API; connect to existing backend APIs.
2. Secure	Apply Authentication, Authorization, and Security policies.
3. Publish	Publish APIs as managed Products for consumers.
4. Consume	Consumers discover and use the published APIs.
5. Monitor	Monitor API usage, traffic, performance, and errors in real time.
6. Govern	Manage API versions, lifecycle, policies, and governance overall.

4. API Management in SAP Integration Suite

SAP API Management is a capability of SAP Integration Suite, used to **publish, secure, monitor,** and **manage** APIs.

5. CPI vs. API Management – what's the difference?

CPI and API Management are complementary capabilities within SAP Integration Suite. They serve **different purposes** but **work together** in most enterprise scenarios.

	SAP Cloud Integration (CPI)	SAP API Management
Focus	Integration / Orchestration	API lifecycle & Governance
Purpose	Connect systems & processes	Expose, secure & manage APIs
1	Connect Systems (Cloud, On-Premise)	Expose APIs (expose backend services and integration flows as APIs)
2	Orchestrate processes (iFlows, routing)	Secure APIs (Authentication, Authorization, OAuth 2.0, API Keys, certificates)
3	Transform & Map (message mapping, MSET, Groovy, content enrichment)	Apply policies (Rate limiting, Quota, Throttling, Transformation, Validation)
4	Adapters (pre-built: HTTP, JDBC, JMS, SFTP, IDoc, Mail, SOAP, OData)	Manage Consumers
5	Business Logic (scripting, routers, content modifiers)	Monitor & Analyze (API traffic, performance, errors)
6	Scheduling (scheduling, batch processing, monitoring of integrations)	API Lifecycle (API versions, lifecycle, portals)

They work together:

```
API Management <-----> CPI Common goal: deliver secure, reliable, and scalable Integrations and APIs.
```

- Use **API Management** to **manage, secure, and govern** your APIs.
- Use **CPI** to **build the integration logic** and **connect to backend systems.**

6. SAP API Management Architecture

API Management acts as a **secure gateway**: API Consumers → Backend System, with API Management sitting in between.

```
Consumers -> SAP API Management -> SAP Cloud Integration -> Backend Systems (Request) Consumers <- Response (JSON / XML) <-
```

How it works, end to end:

- **1.** Consumer calls the API Proxy (HTTPS).
- **2.** API Management applies the Security & Policies.
- **3.** If the request is valid, it's forwarded to CPI (the Target).
- **4.** CPI processes the request and calls the Backend System.
- **5.** Response comes back to CPI.

- **6.** Response policies are applied (by API Management).
- **7.** Response is returned to the Consumer.

7. Real-time example – mobile app requesting Sales Order details

A mobile app requests **Sales Order details** from **SAP S/4HANA**, in real-time, via **SAP API Management.**

```
API Consumer -> SAP API Management -> Backend System (S/4HANA)
```

Real-time flow:

- **1.** User clicks "Get Details."
- **2.** The API call hits the API Proxy.
- **3.** API Management validates the request, applies policies, and routes the request.
- **4.** The request reaches SAP S/4HANA.
- **5.** The response comes back to API Management.
- **6.** The response is returned to the mobile app.

Outcome – topics covered in this introduction

- What is an API
- Why do we need an API
- What is API Management
- Why do we need API Management
- CPI vs. API Management
- SAP API Management Architecture
- How SAP API Management works – end-to-end flow
- Key components of SAP API Management